

Q1. String Character Frequency Counter

CODE

```
s = input().lower()

count = {}

for ch in s:
    if ch != " ":
        if ch in count:
            count[ch] += 1
        else:
            count[ch] = 1

for ch in sorted(count):
    print(ch, "→", count[ch])
```

INPUT

hello world

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics**CODE**

```
s=input().lower()

v="aeiou"
vc={}
cc={}
vow=con=0

for i in s:
    if i.isalpha():
        if i in v:
            vow+=1
            vc[i]=vc.get(i,0)+1
        else:
            con+=1
            cc[i]=cc.get(i,0)+1

print("Vowels:",vow)
print("Consonants:",con)

if vc:
    m=max(vc,key=vc.get)
    print("Most frequent vowel:",m,"(",vc[m],")")
else:
    print("Most frequent vowel: None")

if cc:
    l=min(cc,key=cc.get)
    print("Least frequent consonant:",l,"(",cc[l],")")
else:
    print("Least frequent consonant: None")

if con:
    print("Vowel-to-Consonant Ratio:",round(vow/con,2))
else:
    print("Vowel-to-Consonant Ratio: 0")
```

INPUT

python programming

OUTPUT

```
Vowels: 4
Consonants: 13
Most frequent vowel: o ( 2 )
Least frequent consonant: y ( 1 )
Vowel-to-Consonant Ratio: 0.31
```

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Q3. String Compression and Run-Length Encoding

CODE

```
s=input()

c=""
i=0

while i<len(s):
    ch=s[i]
    n=1
    while i+1<len(s) and s[i]==s[i+1]:
        n+=1
        i+=1
    c+=ch if n==1 else ch+str(n)
    i+=1

if len(c)>=len(s):
    print("Compression not beneficial.")
    print("Original:",s)
else:
    print("Compressed:",c)

    d=""
    i=0
    while i<len(c):
        ch=c[i]
        i+=1
        num=""
        while i<len(c) and c[i].isdigit():
            num+=c[i]
            i+=1
        d+=ch*(int(num) if num else 1)

    print("Decompressed:",d)
```

INPUT

aabbccddddddee

OUTPUT

Compressed: a2b3c2d5e2
Decompressed: aabbccddddddee

Q4. Advanced Text Statistics Analyser

CODE

```
import re
from collections import Counter

text=input()

stop={"is","a","an","the","in","of","and","to","for","on","with"}

words=re.findall(r"[A-Za-z]+",text.lower())
wc=len(words)

sent=max(1,len(re.findall(r"[.!?]",text)))
para=1

avgw=round(sum(len(i) for i in words)/wc,2) if wc else 0
avgs=round(wc/sent,2)

f=Counter(w for w in words if w not in stop)

top=", ".join([f"{k}({v})" for k,v in f.most_common(5)])

def syl(w):
    return max(1,len(re.findall(r"[aeiouy]+",w)))

sy=sum(syl(i) for i in words)
score=round(206.835-1.015*(wc/sent)-84.6*(sy/wc),1)

print("Words:",wc,"| Sentences:",sent,"| Paragraphs:",para)
print("Avg Word Length:",avgw,"| Avg Sentence Length:",avgs)
print("Top 5 Words:",top)
print("Flesch Reading Ease:",score)
```

INPUT

Python is a powerful programming language. It is easy to learn. Python is widely used in data science. Programming with Python is fun.

OUTPUT

Words: 23 | Sentences: 4 | Paragraphs: 1
Avg Word Length: 4.7 | Avg Sentence Length: 5.75
Top 5 Words: python(3), programming(2), powerful(1), language(1), it(1)
Flesch Reading Ease: 53.9